Convection

Exam code: 0625 & 0972

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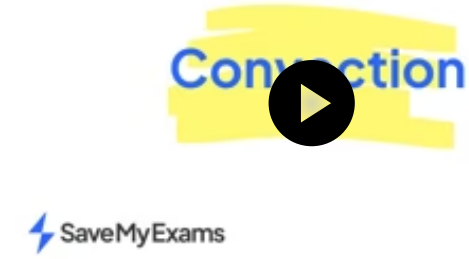

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Help me understand

Convection

- Convection is the main way that heat travels through **liquids and gases**
  - Convection **only** occurs in fluids
  - Convection **cannot** happen in solids

Density & convection

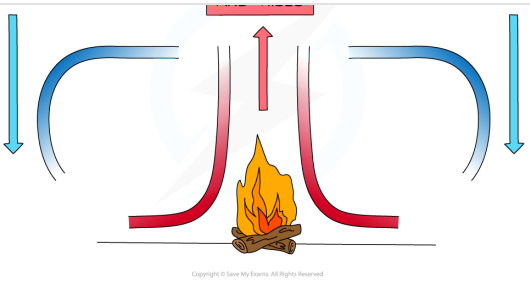
- Descriptions of convection currents always need to refer to changes in temperature causing changes in **density**
- The temperature may fall or rise, both can create a convection current
- **When a liquid (or gas) is heated (for example by a radiator near the floor)**:
- The heated molecules vibrate and push each other apart, **making the liquid/gas expand**
- This makes the hot liquid/gas **less dense** than the surroundings
- The **hot liquid/gas rises**, and the cooler (surrounding) liquid/gas moves in to take its place
- Eventually the hot liquid/gas cools and contracts, increasing in density, and sinks back down again
- The resulting motion is called a **convection current**

Simple convection current diagram

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



When a liquid or gas is heated, it becomes less dense and rises

- **When a liquid (or gas) is cooled (for example by an A.C. unit high up on a wall)**:
  - The molecules lose energy and collide less, **making the liquid/gas contract**
  - This makes the cold liquid/gas **more dense** than the surroundings
  - The **cold liquid/gas falls**, so that warmer, less dense, liquid or gas can move into the space created
  - The warmer liquid/gas gets cooled and also contracts and falls down
  - The resulting motion is called a **convection current**

Demonstrating convection currents

- A simple demonstration of convection in liquids involves taking a beaker of water and placing a few crystals of potassium permanganate in it as shown in the simple convection current diagram below – this compound is purple in water
- When the water is heated around the crystals, the potassium permanganate will dissolve in the heated water and rise along with the warmed water, revealing the convection current
- As the warmed water cools and becomes more dense, it will fall and carry the potassium permanganate with it

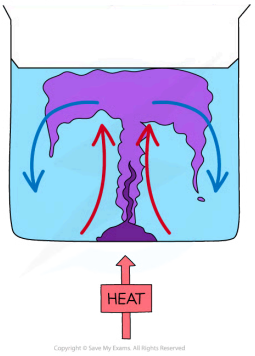
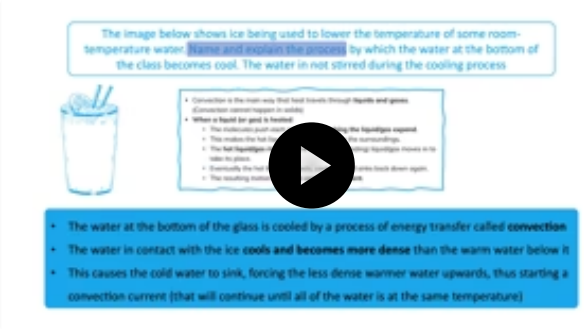


Diagram showing an experiment with potassium permanganate to demonstrate convection

probably be about **convection**.

Heat does not rise – it is the hot gases or liquids which rise due to the change in density when they were heated.



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